Tutorial 2 (One-way ANOVA and Two-way ANOVA)

Part 1

1. Using the data in Question 1 of Tutorial 1 (test_anxiety.csv), fit the model in R using the aov() function and extract the model estimates. Do they match what you obtained with the manual calculations?
2. Based on the estimate of the standard error for differences in treatment means can you calculate the estimated value of σ^2? Hint: use the second formula shown in section 6.4 of the notebook.
3. Using the data in Question 2 of Tutorial 1(remember to use the same seed), fit the model in R using the aov() function and extract the model estimates.
4. For both analyses, briefly interpret the results. Your answer should refer to the research question and explain what the results suggest in practical terms, not just whether the test was significant.
5. In a completely randomized design, 15 experimental units were used in each of three treatments. Complete the following analysis of variance. At a 0.05 level of significance, is there a significant difference between the treatments?

Source of variation	Sum of squares	Df	Mean square	F	p-value
Treatments	1200				
Error					
Total	1800				

6. ANOVA is a procedure intended to compare:
   A. More than two population variances
   B. More than two population means
   C. Two population variances
   D. Two population medians
   E. Two population means

Part 2

Experiment Description

A researcher wants to determine if different study environments impact student performance. Three groups of students, each with 5 participants, were randomly assigned to different study environments:

- Group 1: Quiet Library

- Group 2: Coffee Shop

- Group 3: Home

At the end of a week, students took a standardized test (score out of 100). The scores recorded are as follows:

Coffee Shop	**Home**	**Library**
70	82	78
74	85	80
72	79	75
69	81	82
73	80	77

The data is available in a csv file called: study_environment.csv

7. Identify the following based on the experiment description:
    a. Treatment factor:
    b. Factor levels:
    c. Treatments:
    d. Response:
    e. Experimental units:
    f. Number of replicates:

8. What is the treatment structure and how has randomization been conducted?

9. Based on the treatment structure and randomisation procedure you have identified what type of design has been used here?
10. Write down the population model equation used for analysing data from this type of experiment. Be specific.
11. Write out the separate model equations for each treatment.
12. Now, assuming that the model assumptions have been checked, fit the model in R using the aov() function and extract the model estimates.
13. What is the estimated mean for the Library treatment? Interpret the mean in words.
14. What is the estimated effect of the Home treatment? Interpret the effect in words.
15. Write out the fitted model equations.

16. What is the 95% confidence interval for each of the treatment means? Interpret in words the confidence interval for one of the means.
17. Perform an ANOVA hypothesis in R using the summary() function. Make sure the data is in the correct format (i.e. long format) and that the treatment variable is correctly read in as a factor.
18. What is the total sums of squares?
19. What is the treatment MS and MSE?
20. How was the F-value obtained? Interpret the F-value in words.
21. What is the conclusion of the hypothesis based on the p-value? Give the p-value and interpret it in words.

Part 3

An experiment has been conducted for four treatments. Complete the following analysis of variance table.

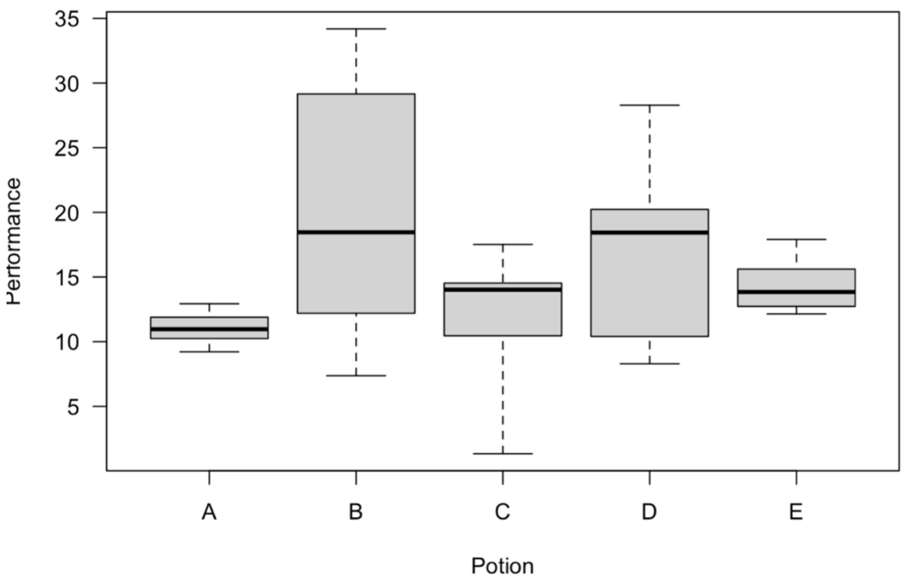
Source of Variation	**Degrees of Freedom**	**Sum of Squares**	**Mean Square**	**F**
Treatments	?	450	?	?
Error	?	?	84.375	
Total	? 19	1800		

Use **α = 0.05** to test for any significant differences.

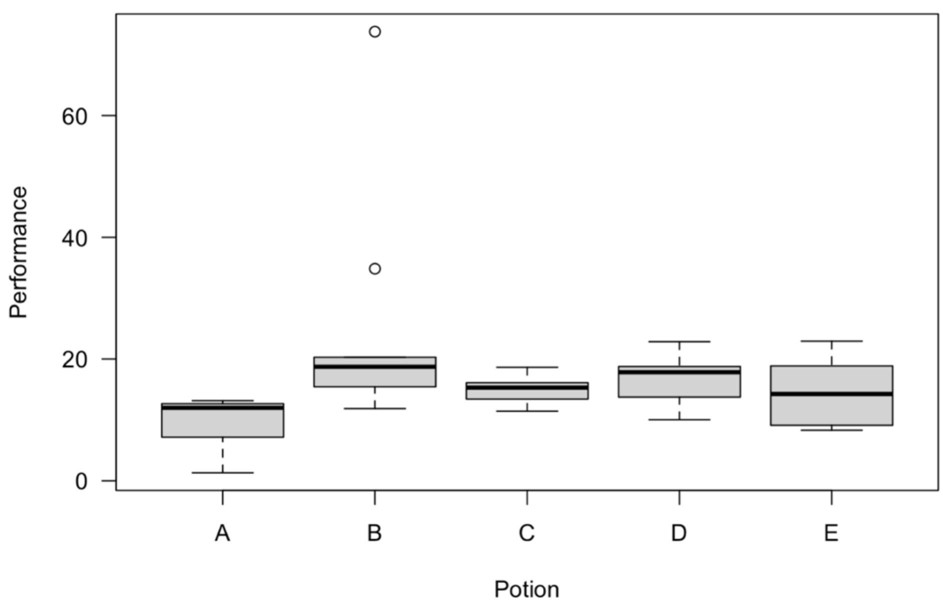
Part 4

Identify in each of the plots below which assumption(s) is being checked and if you think there are any violations.

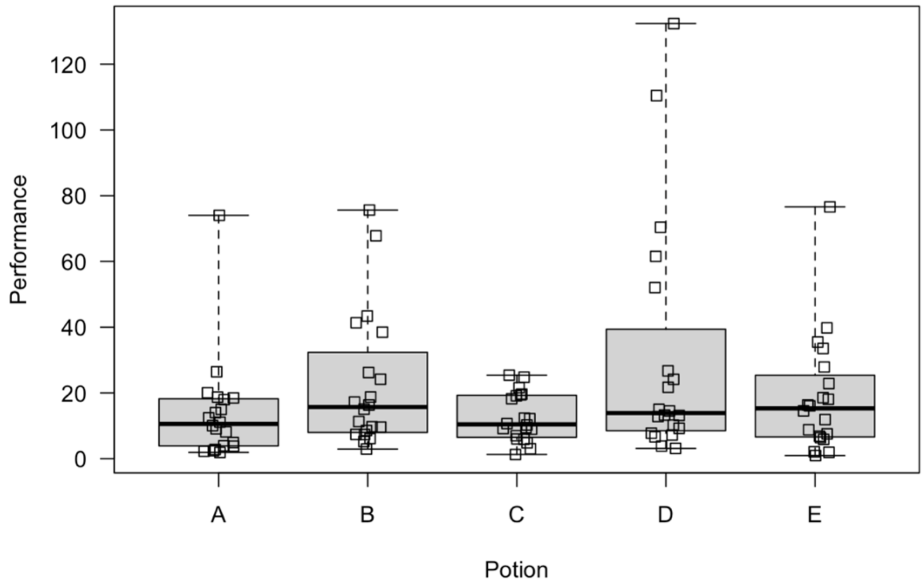
1.


35
30
25
20
15
10
5
Performance
A
B
C
D
E
Potion

2.

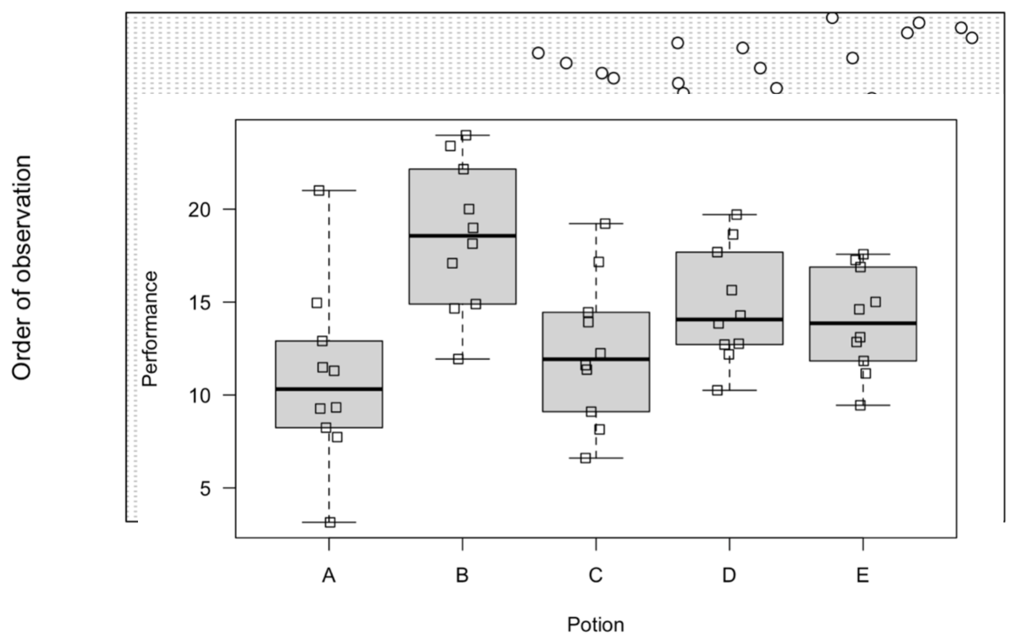

60
40
20
0
Performance
A
B
C
D
E
Potion

3.



Performance
0
20
40
60
80
100
120
A
B
C
D
E
Potion

4.



Order of observation
Performance
5
10
15
20
A
B
C
D
E
Potion

5.

Part 5

A taste-testing experiment has been designed so that four brands A to D of Columbian coffee are to be rated by nine experts. The experimenters are interested in the differences in mean ratings between the four brands. The tasting sequence of the four brands is randomly determined for each of the nine expert tasters until a rating on a 7-point scale (1 = extremely unpleasing, 7 = extremely pleasing) is given for each of four characteristics: taste, aroma, richness, and acidity. The data are the total rating, summed over all four characteristics.

Blocks (Experts)	A	B	C	D	Totals
E.B.	24	26	25	22	97
N.B.	27	27	26	24	104
M.D.	19	22	20	16	77
M.H.	24	27	25	23	99
B.J.	22	25	22	21	90
R.J.	26	27	24	24	101
B.K.	27	26	22	23	98
B.M.	25	27	24	21	97
J.S.	22	23	20	19	84
Totals	216	230	208	193	847

1. Identify the following based on the experiment description:
    a. Treatment factor:
    b. Factor levels:
    c. Treatments:
    d. Response:
    e. Experimental units:
    f. Number of replicates:

2. Why are the experts blocks?
3. What is the treatment structure and how has randomisation been conducted?
4. Based on the treatment structure and randomisation procedure you have identified what type of design has been used here?
5. Write down the population model equation used for analysing data from this type of experiment. Be specific.
6. Calculate the following by hand and note down the R code you could use to calculate the first three:
    a. Overall mean
    b. Treatment means and note down the formula for calculating their standard errors.
    c. Treatment effects and note down the formula for calculating their standard errors
7. What are the assumptions of this model?
8. What does the assumption of additivity between blocks and treatments mean? Put in context of the experiment.
9. Assuming you have checked the assumptions and they are reasonably met. Conduct the ANOVA hypothesis by hand and summarise the results in an ANOVA table.
10. What conclusion can you draw about the treatment factor based on the result you obtained? Provide a p-value to support your conclusion.
11. Was blocking effective in this experiment? Motivate your answer.

Part 5 – Extra Practice

1. An experiment has been conducted for four treatments with eight blocks. Complete the following analysis of variance table.

Source of variation	**SS**	**DF**	**MS**	**F**

Treatment	900			
Blocks	400			
Error				
Total	1800			

Use α = 0.05 to test for any significant differences.

2. A car dealer, *AfricaDrive*, conducted a test to determine if the time in minutes needed to complete a minor engine tune-up depends on whether a computerized engine analyser or an electronic analyser is used. Because tune-up time varies among compact, intermediate and full-sized cars, the three types of cars were used as blocks in the experiment. The data obtained follow.

Car	Computerized	Electronic
Compact	50	42
Intermediate	55	44
Full-sized	63	46

Use α = 0.05 to test for any significant differences.

3. To test for any significant difference in the number of hours between breakdowns for four machines, the following data were obtained.

Machine 1	Machine 2	Machine 3	Machine 4
6.4	8.7	11.1	9.9
7.8	7.4	10.3	12.8
5.3	9.4	9.7	12.1
7.4	10.1	10.3	10.8
8.4	9.2	9.2	11.3
7.3	9.8	8.8	11.5

At the α = 0.05 level of significance, what is the difference, if any, in the population mean times among the four machines?

4. A randomized block design (RBD) produced the following data with five blocks and three treatments:

Block	**1**	**2**	**3**
1	25	27	25
2	19	18	17
3	15	20	16
4	23	27	20
5	30	31	28

Perform an ANOVA and explain your conclusions.

5. An RBD with 4 treatments and 5 blocks produced the following:

SS Total = 1951
SS Treatments = 349
SS Error = 188

It follows that SS Blocks is equal to:

A. 1414
B. 537
C. 1763
D. 1602
E. None of the above

6. We introduce blocks into the design of an experiment for one main purpose. This is:

A. To reduce between block variation
B. To maximize within treatment variation
C. To reduce between treatment variation
D. To reduce error variation
E. To maximize within block variation

7. In an RBD with *k* treatments and *b* blocks, the degrees of freedom for error are:
A. k − 1
B. b − 1
C. bk

D. (b − 1)(k − 1)
E. bk – 1

8. Suppose we want to compare starting salaries of graduates from 15 South African universities. We take a random sample of 25 graduates from each university and perform one-way ANOVA. What are the degrees of freedom for the test statistic?

   A. 15 and 25
   B. 14 and 360
   C. 360 and 14
   D. 25 and 15
   E. 14 and 24

9. In an RBD the null hypothesis of no block differences is rejected if:

   A. MS Blocks is much smaller than MS Error
   B. MS Treatments is much larger than MS Blocks
   C. MS Treatments is much larger than MS Error
   D. MS Blocks is much larger than MS Treatments
   E. MS Blocks is much larger than MS Error